

Qatar-1b

R-0291

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_171025 · created 2026-07-17T17:56:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458051.7338 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.012
Transit depth (Rp/Rs)^2	2.68 +/- 0.38 [%]
Orbital Inclination (inc)	83.76 +/- 0.61
Airmass coefficient 1 (a1)	0.968 +/- 0.012
Airmass coefficient 2 (a2)	0.0184 +/- 0.0058
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#2 - [368, 92]
Optimal Aperture	3.22
Optimal Annulus	12.32
Transit Duration (day)	0.0688 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.012 vs 0.1463 (deviation 1.35σ)
Duration fit vs published	0.0688 d vs 0.0692 d (deviation 0.1σ)
Mid-time O–C	-0.9 min
Depth SNR	12.5
Residual scatter	1.24 %

Light curve

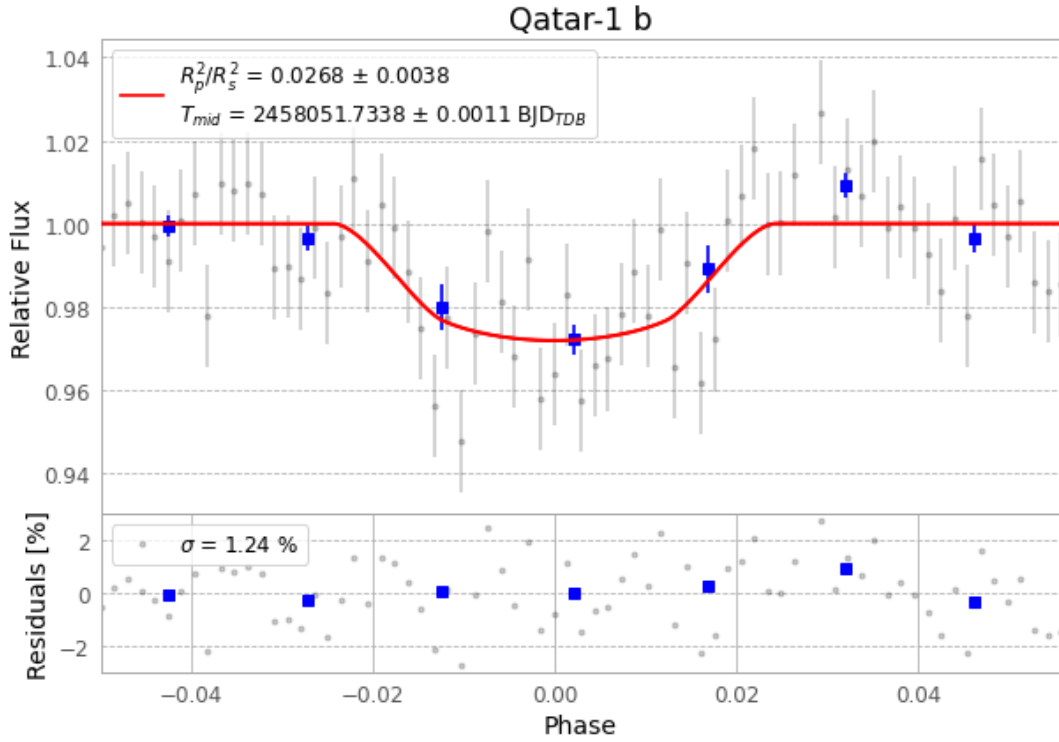


Figure 1 — FinalLightCurve_Qatar-1 b_2017-10-24.png (SHA-256 37bce1e27b4b430c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 352], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c88dc110f2c9bdf7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

73 FITS frames (48 MB), Qatar-1171025035112.FITS → Qatar-1171025072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-171025.log (ea9acd4f1d53...), FinalLightCurve_Qatar-1 b_2017-10-24.png (37bce1e27b4b...), FinalLightCurve_Qatar-1 b_2017-10-24.csv (42c1db9a57b0...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 17:56Z.